

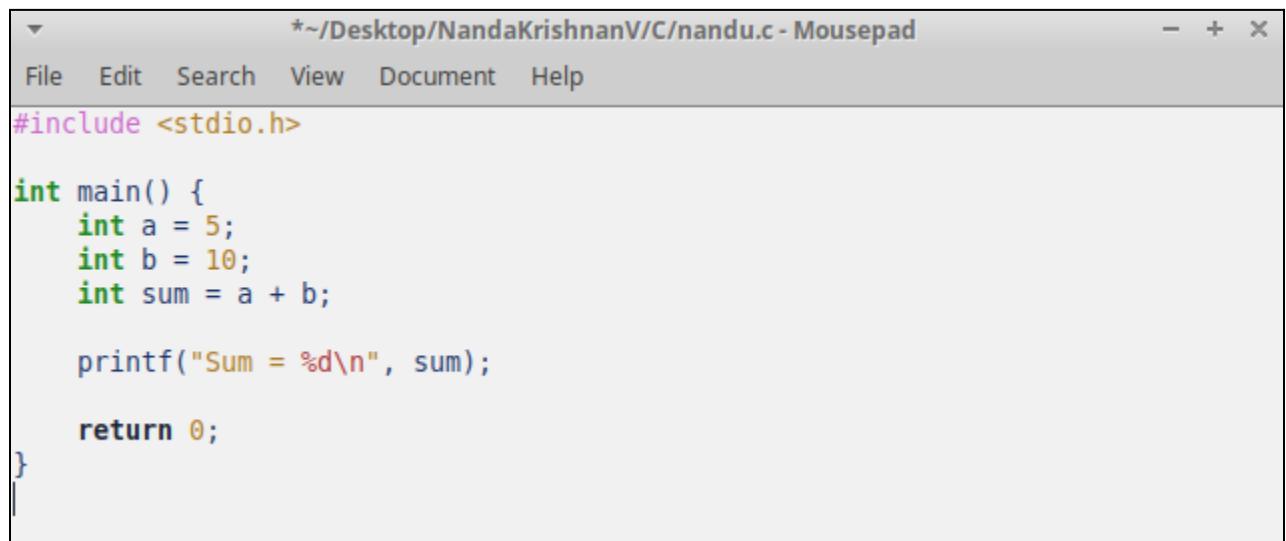
Malware Analysis A1

Objective:

Write a basic C program and compile it to generate both .exe and .elf executable files. Using the file commands, examine the basic details of these files. Next, utilize the xxd command to analyze their hexadecimal content and identify the file type. Generate cryptographic hashes (MD5 and SHA-256) for these executables and finally verify the files using the VirusTotal service.

Source Code (nandu.c):

A simple C program was created to perform a basic arithmetic operation (addition) and print the result to the console.

A screenshot of a code editor window titled '*~/Desktop/NandaKrishnanV/C/nandu.c - Mousepad'. The window has a menu bar with 'File', 'Edit', 'Search', 'View', 'Document', and 'Help'. The code is written in C and is as follows:

```
#include <stdio.h>

int main() {
    int a = 5;
    int b = 10;
    int sum = a + b;

    printf("Sum = %d\n", sum);

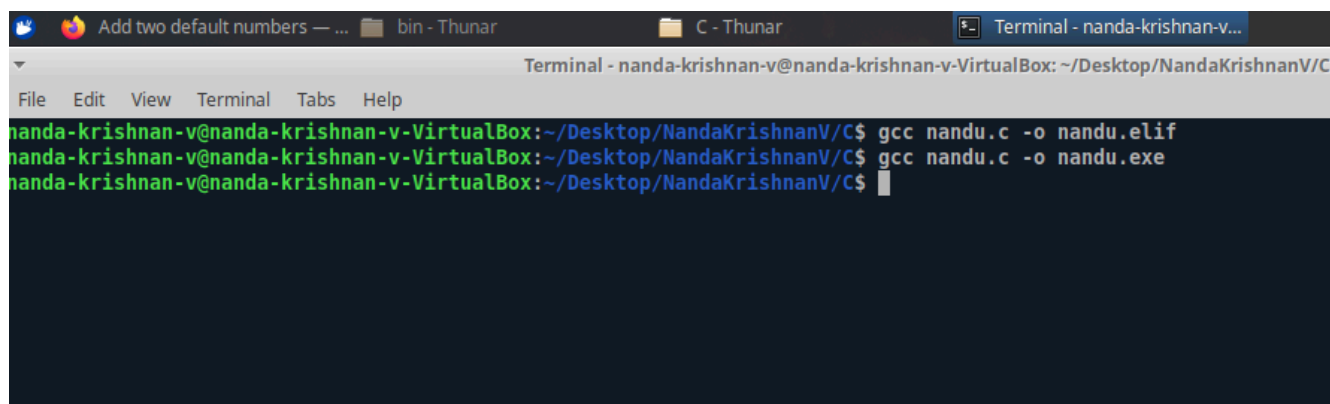
    return 0;
}
```

Compilation:

The C source code was compiled using the GNU C Compiler (gcc) on a Linux-based system. Two separate output files, nandu.elf and nandu.exe, were intentionally created to demonstrate that the file extension does not determine the file type on a Linux system.

Commands Used:

```
gcc nandu.c -o nandu.elf
gcc nandu.c -o nandu.exe
```

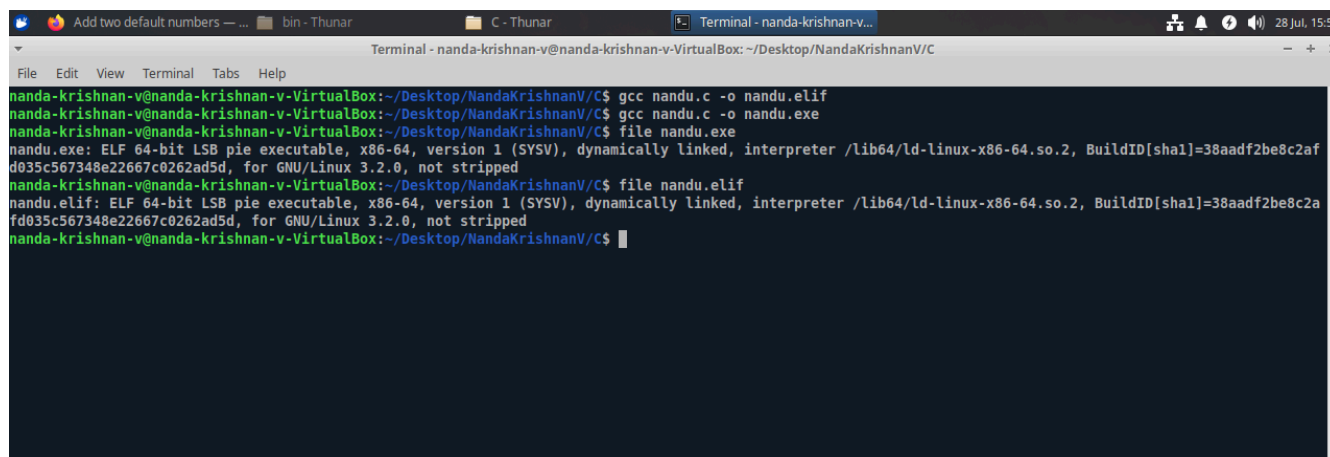


```
Terminal - nanda-krishnan-v@nanda-krishnan-v-VirtualBox: ~/Desktop/NandaKrishnanV/C
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ gcc nandu.c -o nandu.elf
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ gcc nandu.c -o nandu.exe
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$
```

Static Analysis

File Identification

- The **file** command revealed that both nandu.exe and nandu.elf are 64-bit ELF (Executable and Linkable Format) files. This is the standard executable file format for Linux and other Unix-like operating systems.



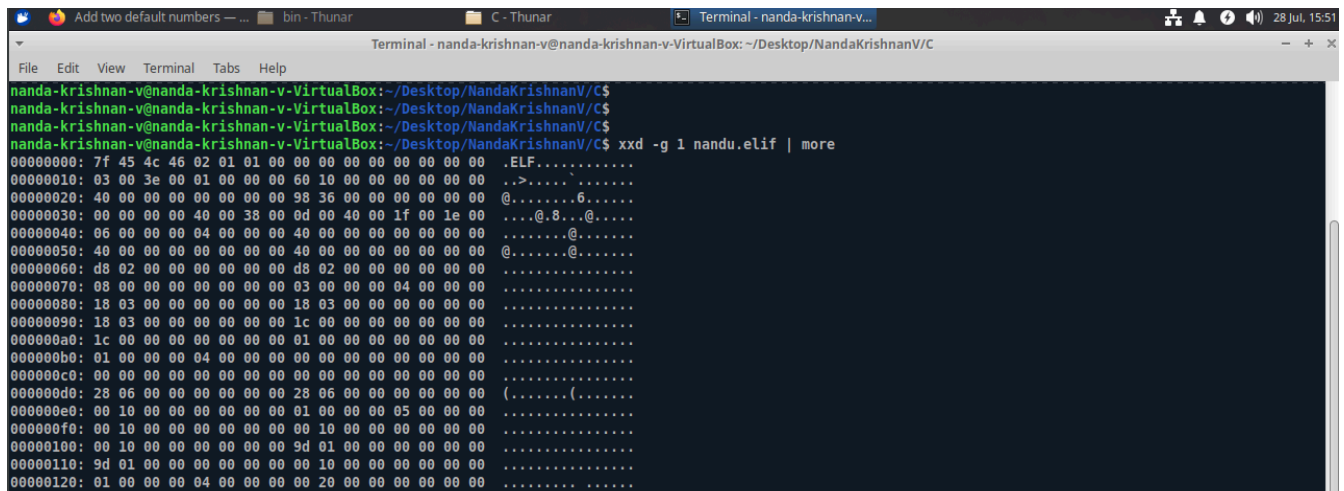
```
Terminal - nanda-krishnan-v@nanda-krishnan-v-VirtualBox: ~/Desktop/NandaKrishnanV/C
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ gcc nandu.c -o nandu.elf
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ gcc nandu.c -o nandu.exe
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ file nandu.exe
nandu.exe: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=38aadf2be8c2afd035c567348e22667c0262ad5d, for GNU/Linux 3.2.0, not stripped
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ file nandu.elf
nandu.elf: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=38aadf2be8c2afd035c567348e22667c0262ad5d, for GNU/Linux 3.2.0, not stripped
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$
```

Hexadecimal Analysis

The xxd command was used to view the raw hexadecimal representation of nandu.elf. A hex dump allows us to see the file's byte-level structure.

- The first few bytes 7f 45 4c 46 are particularly important.
- 0x7f is a constant.

- 45 4c 46 are the ASCII codes for 'E', 'L', 'F'.



```
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$  
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$  
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$  
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ xxd -g 1 nandu.elif | more  
00000000: 7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00 .ELF.....  
00000010: 03 00 3e 00 01 00 00 00 60 10 00 00 00 00 00 00 ..>.....  
00000020: 40 00 00 00 00 00 00 00 98 36 00 00 00 00 00 00 @.....6.....  
00000030: 00 00 00 00 40 00 38 00 0d 00 40 00 1f 00 1e 00 ....@.8...@.....  
00000040: 06 00 00 00 04 00 00 00 40 00 00 00 00 00 00 00 .....@.....  
00000050: 40 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 @.....@.....  
00000060: d8 02 00 00 00 00 00 00 d8 02 00 00 00 00 00 00 .....  
00000070: 08 00 00 00 00 00 00 00 03 00 00 00 04 00 00 00 .....  
00000080: 18 03 00 00 00 00 00 00 18 03 00 00 00 00 00 00 .....  
00000090: 18 03 00 00 00 00 00 00 1c 00 00 00 00 00 00 00 .....  
000000a0: 1c 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 .....  
000000b0: 01 00 00 00 04 00 00 00 00 00 00 00 00 00 00 00 .....  
000000c0: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....  
000000d0: 28 06 00 00 00 00 00 00 28 06 00 00 00 00 00 00 (.....(  
000000e0: 00 10 00 00 00 00 00 00 01 00 00 00 05 00 00 00 .....  
000000f0: 00 10 00 00 00 00 00 00 00 10 00 00 00 00 00 00 .....  
00000100: 00 10 00 00 00 00 00 00 9d 01 00 00 00 00 00 00 .....  
00000110: 9d 01 00 00 00 00 00 00 00 10 00 00 00 00 00 00 .....  
00000120: 01 00 00 00 04 00 00 00 00 20 00 00 00 00 00 00 .....
```

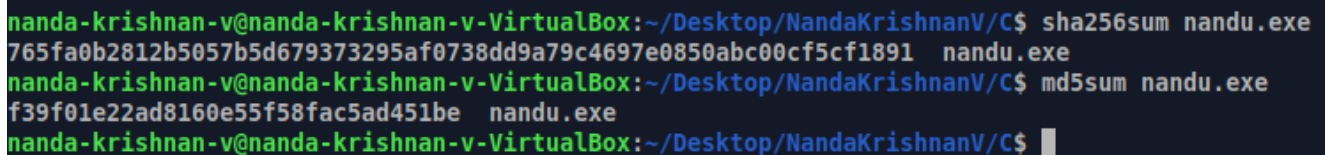
Hash-Based Identification:

This is to create a unique, fixed-length "fingerprint" of the file. Any change to the file, no matter how small, will result in a completely different hash. This is crucial for tracking specific malware samples.

Hashes Generated for nandu.exe:

MD5: f39f01e22ad8160e55f58fac5ad451be

SHA256: 765fa0b2812b5057b5d679373295af0738dd9a79c4697e0850abc00cf5cf1891



```
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ sha256sum nandu.exe  
765fa0b2812b5057b5d679373295af0738dd9a79c4697e0850abc00cf5cf1891 nandu.exe  
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$ md5sum nandu.exe  
f39f01e22ad8160e55f58fac5ad451be nandu.exe  
nanda-krishnan-v@nanda-krishnan-v-VirtualBox:~/Desktop/NandaKrishnanV/C$
```

Verification on [VirusTotal](#):

I uploaded the nandu.exe file to VirusTotal, which scanned the file using 65+ antivirus engines and resulted in 0/65 engines flagging the file as malicious. This confirms the compiled executable is clean and safe.

765fa0b2812b5057b5d679373295af0738dd9a79c4697e0850abc00cf5cf1891

Community Score: 0 / 65

No security vendors flagged this file as malicious

765fa0b2812b5057b5d679373295af0738dd9a79c4697e0850abc00cf5cf1891

nandu.exe

Size: 15.59 KB

Last Analysis Date: a moment ago

elf 64bits shared-lib

Reanalyze Similar More

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Security vendors' analysis

Security vendors' analysis		Do you want to automate checks?	
Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
AliCloud	Undetected	ALYac	Undetected
Antiy-AVL	Undetected	Arcabit	Undetected
Avast	Undetected	Avast-Mobile	Undetected
AVG	Undetected	Avira (no cloud)	Undetected

Conclusion

The analysis confirms that the compiled C program, nandu.exe, is a benign 64-bit ELF executable. The file properties are consistent with a standard, non-malicious program compiled on a Linux system. The clean bill of health from VirusTotal, with a 0/65 detection rate, further substantiates this conclusion. This exercise successfully establishes a baseline for identifying the characteristics of a safe file, which can be contrasted with the characteristics of malicious files in future analyses